

Week 3: Data pipeline and first results

Code: github.com/ltsabhirajurs/Trust_DTI (src/ingest, tests, reports/stage0) · Full data report: docs/stage0_report.md

BUILT

Verified downloads, extraction for both databases, profiling, overlap and source analysis, 10 tests.

FOUND

5.06M drug–protein pairs on 5,738 proteins. The databases copy each other. Screening data distorts potency per family.

NEXT

Lock the dataset rules (endpoints, thresholds, inactive labels, minimum data per protein, time split) from these numbers.

1 What we built

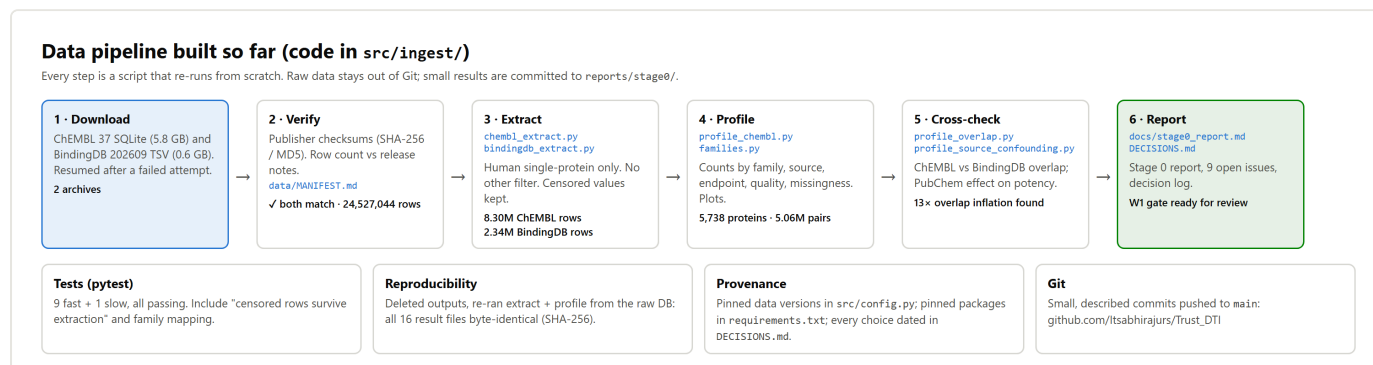


Figure 1. Pipeline steps, the script behind each, and the checks around them.

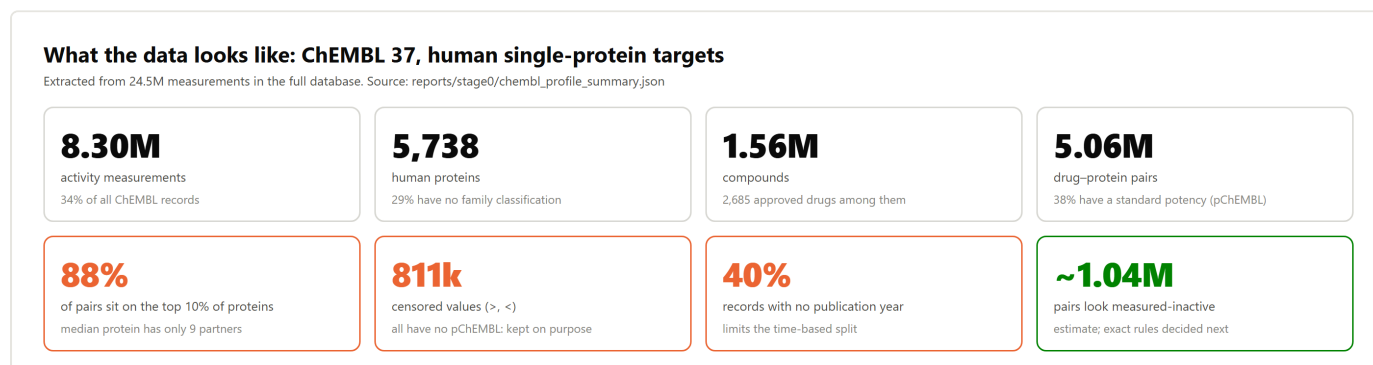


Figure 2. The dataset at a glance (ChEMBL 37, human single-protein targets).

2 What the data showed

Uneven coverage. Enzymes dominate, and data is concentrated on a few proteins: the top 10% of proteins hold 88% of all pairs, while half have fewer than 10 partners. This affects how we split data for "new protein" testing.

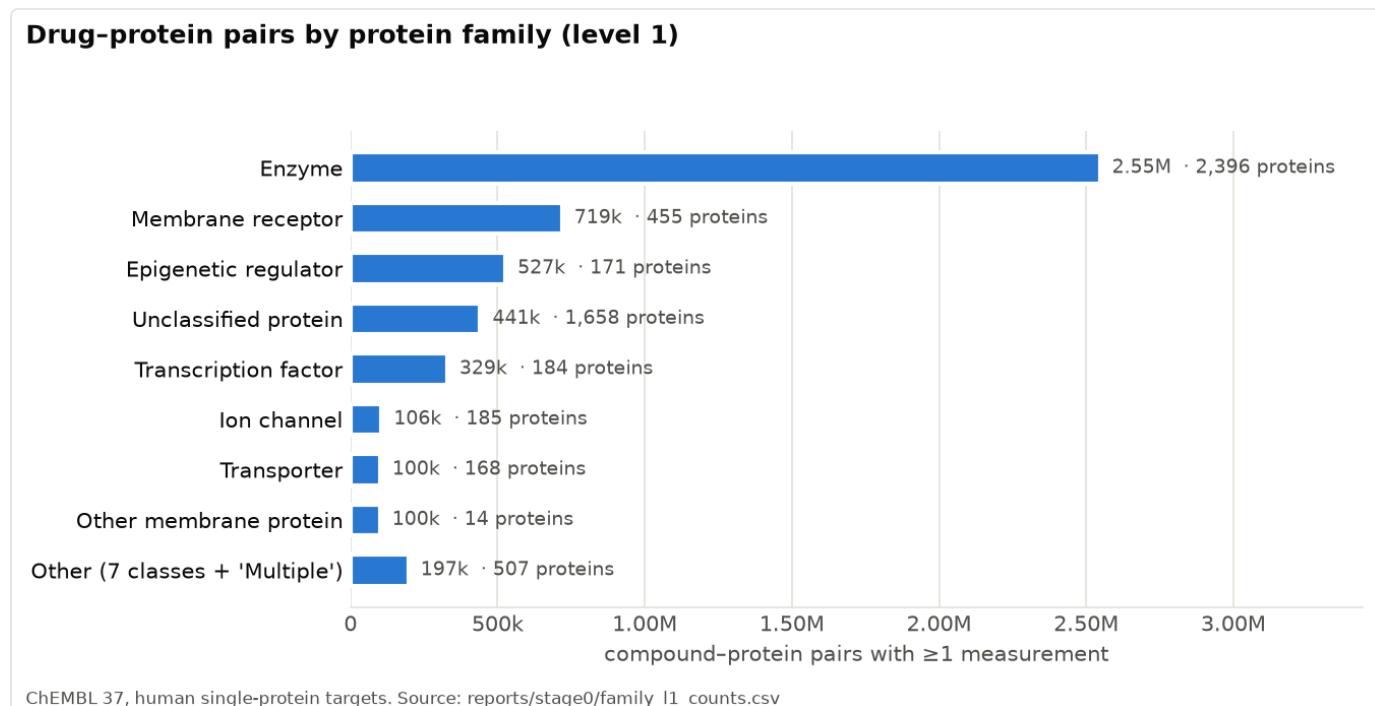


Figure 3. Pairs and proteins per family. 1,658 proteins (29%) have no family classification but only sparse data.

Mixed origins. A third of records are PubChem screening results, which have no publication year and so cannot be used in a time-based split.

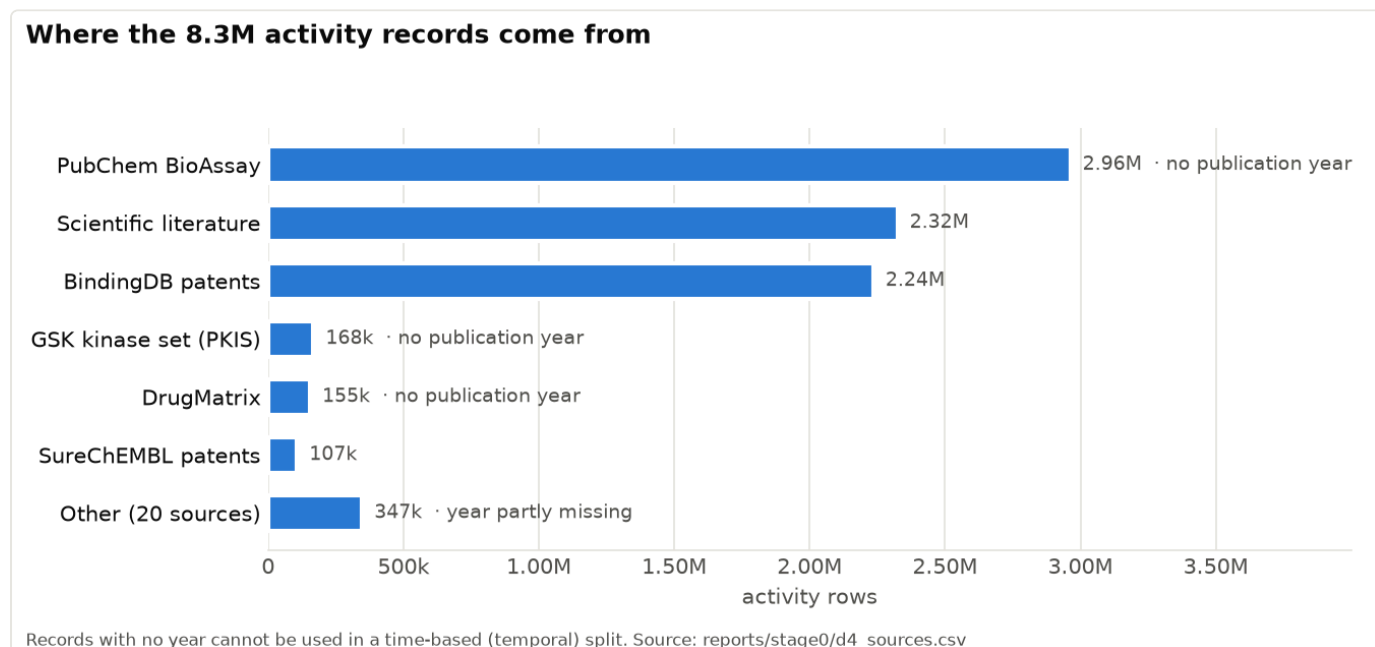
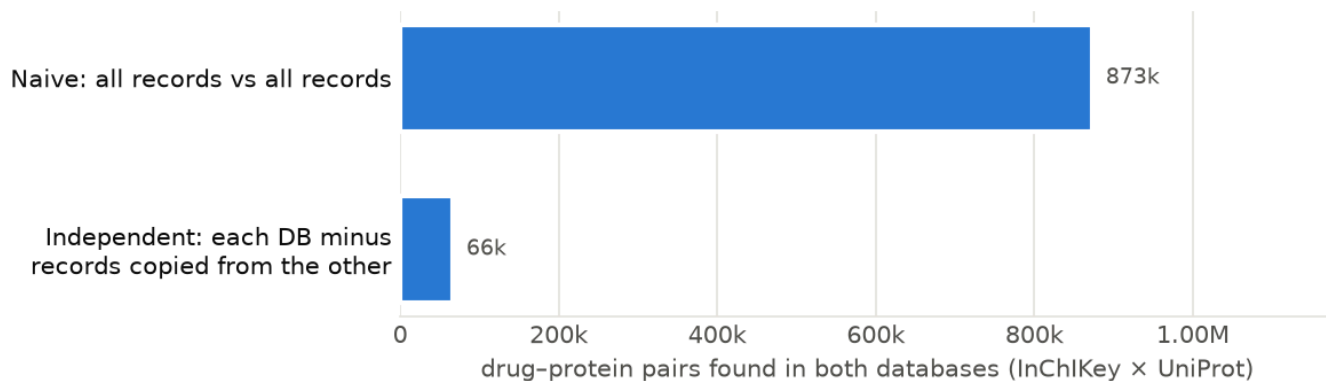


Figure 4. Records by source.

3 Two findings that change the plan

(a) The second database is mostly a copy. 52% of BindingDB's human records come from ChEMBL, and ChEMBL imports BindingDB's patent data. Counted naively the overlap looks large; on independent records only it is 13× smaller. BindingDB would add about 8% new pairs, mostly patents (likely newer than ChEMBL 37, which is still to be checked). **Recommendation:** use ChEMBL only for the first network, and check whether those patents could strengthen the time-based test.

ChEMBL n BindingDB shared pairs: the naive count is ~13× inflated

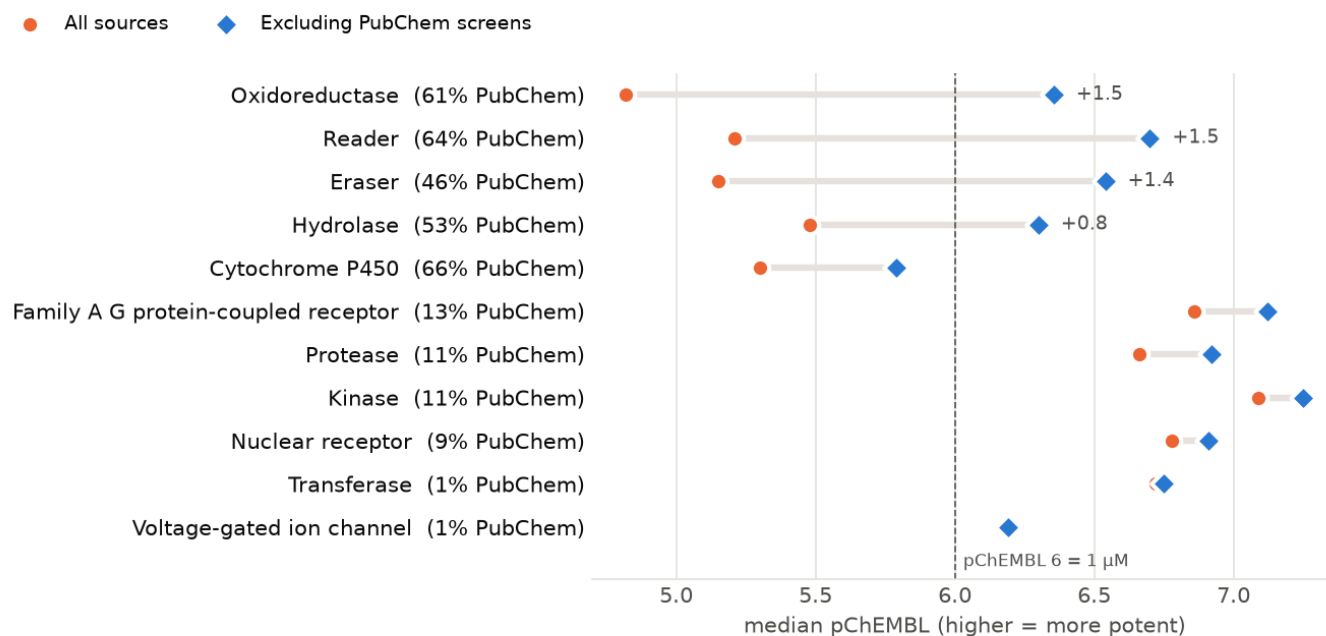


52% of BindingDB human rows are copied from ChEMBL; ChEMBL imports BindingDB patent data. Source: reports/stage0/d3_overlap.csv

Figure 5. Shared drug-protein pairs, naive vs independent.

(b) Screening data distorts family potency. 80% of the low-potency spike at pChEMBL ≈ 4.5 comes from PubChem screens. For oxidoreductases the median moves from 4.8 to 6.4 when screens are excluded. A per-family activity threshold must therefore control for data source, or it will treat screening artefacts as biology.

Screening data drags some families' measured potency down



Families with $\geq 50k$ pChEMBL values. A per-family threshold must control for data source. Source: reports/stage0/pchembl_source_confounding.csv

Figure 6. Median potency per family, with and without PubChem screening data.

4 Trial and error

Trial and error: what went wrong, how we caught it, what we changed

Each of these would have silently changed a result if it had not been checked.

PROBLEM	HOW IT WAS CAUGHT	FIX	LESSON
ChEMBL download failed: a 265 MB partial file of 5.8 GB looked usable	gzip: unexpected end of file on inspection	Resumed with <code>curl -C -</code> ; accepted only after the SHA-256 matched EBI's published value	Trust checksums, not file names or sizes
Repo inside OneDrive , with a 30 GB database about to sync	Environment review before starting	Moved the working copy to <code>C:\Users\91701\code</code> ; OneDrive copy kept as backup	Keep big data out of synced folders
BindingDB: 5 rows "missing" after parsing 9 GB	Reconciled line count (3,237,052) with rows read (3,237,047)	Found 5 blank lines + 1 corrupt NUL line in the publisher's file; confirmed no data lost	Always reconcile counts end to end
Notebook over-counted censored rows	Relation counts didn't add up (859 blank relations)	Replaced <code>!=</code> with an explicit operator list	Missing <code>≠</code> "not equal"
Unexpected operators <code>>></code> and <code><<</code> (70 rows)	Full relation table in profiling	Added to the censored set; unit test updated	Profile before assuming a field's values
ChEMBL-BindingDB overlap looked huge (873k shared pairs)	BindingDB's source column: 52% of its rows come from ChEMBL	Measured overlap on independent records only: 66k (13× smaller)	Check where "second sources" got their data
Family potency looked very different (oxidoreductase median 4.8)	A spike at pChEMBL = 4.5 in the histogram	80% of the spike is PubChem screening; without it the median is 6.4. Flagged for threshold design	Separate data-source effects from biology
A test took 58 minutes	Timed test run	Marked the full-database test <code>slow</code> ; the default suite runs in 2 s	Keep the everyday test loop fast

Figure 7. Problems found while building, how each was caught, and what changed.

5 Quality checks

- **Checksums:** ChEMBL SHA-256 and BindingDB MD5 both match the publishers' values. The database row count (24,527,044) matches the release notes.
- **Tests:** 10 automated tests, all passing. They cover censored-value parsing, the family mapping, "censored rows survive extraction" (checked against a direct SQL count) and a full-database row check.
- **Reproducibility:** we deleted the outputs and re-ran extraction and profiling from the raw database. All 16 result files came out byte-identical.

6 Open questions for next week

1. Which endpoints count: IC50/Ki/Kd/EC50 only, or also screening "Potency" (2.69M rows)?
2. Activity thresholds per family that control for data source (finding b).
3. Rules for measured-inactive labels (~1.04M candidate pairs); "inconclusive" stays UNTESTED.
4. Time-split cutoff, given that 40% of rows have no year.
5. Minimum data per protein for a meaningful "new protein" test (2,883 proteins have <10 partners).
6. Confidence floor (only 8 or 9 occur); handling of duplicates and curator-flagged values; BindingDB decision.

7 How to reproduce

```
pip install -r requirements.txt
python -m src.ingest.chembl_extract          # needs data/raw/chembl_37 (see data/MANIFEST.md)
python -m src.ingest.bindingdb_extract
python -m src.ingest.profile_chembl
python -m src.ingest.profile_overlap
python -m src.ingest.profile_source_confounding
pytest                                     # fast tests; pytest -m slow for the full-database check
```

Supporting files in this folder: W3_01–W3_09 (figures; W3_07 interaction skew and W3_08 potency distribution are in the folder only), references.md, WEEK03_FORM_ENTRIES.md. Figures are built from committed data by `docs/weekly_reports/_build/`.